# 1.8 Integration

*Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics*

Integration is the reverse process of differentiation. Where differentiation tells you how a quantity is changing, integration lets you recover the quantity from its rate of change — or, geometrically, compute the area under a curve. The two operations are bound together by the *fundamental theorem of calculus*, one of the most important results in mathematics. This chapter develops the techniques of indefinite and definite integration — power rule, standard integrals, substitution, integration by parts, partial fractions, and trigonometric identities — and uses them to find areas, volumes of revolution, and solutions to first-order separable differential equations. Every applied problem involving accumulation, total change, work, or volume ultimately rests on integration.

## 1. Integration as the Reverse of Differentiation

If differentiation takes a function $F(x)$ and produces its derivative $f(x) = F'(x)$, then integration runs the process backwards: given $f(x)$, find a function whose derivative is $f(x)$. Such a function is called an *antiderivative* of $f$. The essential subtlety is that antiderivatives are not unique — adding any constant to an antiderivative still yields an antiderivative, because the derivative of a constant is zero.

**Indefinite integral (antiderivative).** An *antiderivative* of $f(x)$ is any function $F(x)$ with $F'(x) = f(x)$. The set of all antiderivatives is denoted \[ \int f(x)\, dx = F(x) + C, \] where $C$ is an arbitrary constant — the *constant of integration*. The symbol $\int$ is an elongated "S" (from "sum"); $dx$ indicates the variable of integration.

Why the constant? If $F'(x) = f(x)$ and $G(x) = F(x) + C$ for any constant $C$, then $G'(x) = F'(x) + 0 = f(x)$ as well. So $F$ and $G$ are equally valid antiderivatives. Conversely, any two antiderivatives of $f$ differ by a constant (this is a consequence of the mean value theorem). The "$+C$" captures this ambiguity.

### Worked Example 1 — Why the constant matters

```
1.  Consider f(x) = 2x.
2.  F(x) = x²        has derivative 2x ✓
3.  G(x) = x² + 7    has derivative 2x ✓
4.  H(x) = x² − 3.4  has derivative 2x ✓
5.  All three are antiderivatives of 2x — they differ only by a constant.
6.  We capture all of them at once by writing:
7.      ∫ 2x dx = x² + C.

// The +C is mandatory in every indefinite integral. Losing it loses a mark.
```

**Exam tip — always write +C for indefinite integrals.** OCR mark schemes list the constant of integration as a standalone awarded mark on almost every indefinite-integral question. Forgetting it is the single most common error at A-level. The only place you don't need $+C$ is in a *definite* integral, because the constant cancels when you subtract $F(a)$ from $F(b)$.

## 2. The Power Rule for Integration

Reversing the power rule $\frac{d}{dx}(x^{n+1}) = (n+1)x^n$ gives the integration power rule. Divide by $n+1$ to isolate $x^n$:

**Power rule for integration.** For any real $n \neq -1$, \[ \int x^n\, dx = \frac{x^{n+1}}{n+1} + C. \] In words: *add one to the power, divide by the new power*. The exclusion $n = -1$ is needed because division by zero would occur. The case $n = -1$ is handled by the logarithm: \[ \int \frac{1}{x}\, dx = \ln|x| + C. \] The absolute value is essential — $\ln$ is only defined for positive arguments, but $1/x$ is defined for all $x \neq 0$.

The power rule extends term-by-term via linearity: for constants $a, b$ and integrable functions $f, g$,

$$\int \big[af(x) + bg(x)\big]\, dx = a\int f(x)\, dx + b\int g(x)\, dx.$$

### Worked Example 2 — Polynomial with mixed powers

Find $\int (3x^2 - 4x + 1)\, dx$.

```
1.  Integrate term by term using the power rule:
2.     ∫ 3x² dx = 3 · (x³/3) = x³.
3.     ∫ −4x dx = −4 · (x²/2) = −2x².
4.     ∫ 1 dx  = x           (since 1 = x⁰, add 1 to get x¹/1.)
5.  Combine and add the constant:
6.     ∫ (3x² − 4x + 1) dx = x³ − 2x² + x + C.
7.  Check by differentiating: d/dx (x³ − 2x² + x + C) = 3x² − 4x + 1 ✓.

// Always verify by differentiating — it catches arithmetic errors in seconds.
```

## 3. Standard Integrals

Reversing the standard derivatives gives a short list of integrals that must be memorised. Each is just a differentiation rule read backwards.

| Function $f(x)$ | Integral $\int f(x)\, dx$ | Notes |
|---|---|---|
| $x^n \ (n \neq -1)$ | $\dfrac{x^{n+1}}{n+1} + C$ | Any real $n$ except $-1$. |
| $1/x$ | $\ln\lvert x\rvert + C$ | Absolute value for $x < 0$. |
| $\sin x$ | $-\cos x + C$ | Minus sign — easily missed. |
| $\cos x$ | $\sin x + C$ | No minus sign. |
| $\sec^2 x$ | $\tan x + C$ | Reverses $(\tan x)' = \sec^2 x$. |
| $e^x$ | $e^x + C$ | Unique: same as function. |
| $e^{kx}$ | $\dfrac{1}{k} e^{kx} + C$ | Divide by the inner coefficient. |

| | | |
|---|---|---|
| $\sin(kx)$ | $-\frac{1}{k}\cos(kx)+C$ | Chain-rule reversal. |
| $\cos(kx)$ | $\frac{1}{k}\sin(kx)+C$ | Chain-rule reversal. |
| $\frac{1}{ax+b}$ | $\frac{1}{a}\ln\lvert ax+b\rvert+C$ | Linear inner — divide by $a$. |

**Exam tip — watch for the sign on sin and cos.** It is easy to mix up which of $\sin$ and $\cos$ picks up a minus sign on integration. The safest rule: *differentiate your answer to check*. For $\int \sin x\,dx = -\cos x + C$: differentiate $-\cos x$ to get $-(-\sin x) = \sin x$ ✓. For $\int \cos x\,dx = \sin x + C$: differentiate $\sin x$ to get $\cos x$ ✓. A five-second check prevents a lost mark.

## 4. Definite Integrals and the Fundamental Theorem

An indefinite integral produces a family of antiderivatives. A *definite* integral produces a number — the net change of an antiderivative between two specified limits, equivalently the (signed) area between the curve and the x-axis.

**Definite integral.** If $F$ is an antiderivative of $f$ on $[a,b]$, then the definite integral of $f$ from $a$ to $b$ is \[ \int_a^b f(x)\, dx = \Big[F(x)\Big]_a^b = F(b) - F(a). \] The numbers $a$ and $b$ are the lower and upper *limits of integration*. No constant of integration is written — it would cancel in $F(b)+C-(F(a)+C)$.

**Fundamental theorem of calculus.** If $f$ is continuous on $[a,b]$, then $F(x)=\int_a^x f(t)\,dt$ is differentiable on $(a,b)$ and $F'(x)=f(x)$. Consequently, if $F$ is any antiderivative of $f$ on $[a,b]$, then \[ \int_a^b f(x)\, dx = F(b) - F(a). \] This theorem ties the two branches of calculus together: the integral is the inverse of the derivative.

Useful properties of definite integrals:

- **Swap limits, change sign:** $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$.
- **Adjacent interval additivity:** $\int_a^b f\,dx + \int_b^c f\,dx = \int_a^c f\,dx$.
- **Zero interval:** $\int_a^a f(x)\,dx = 0$.

- **Linearity:** $\int_a^b [\alpha f + \beta g]\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx$.

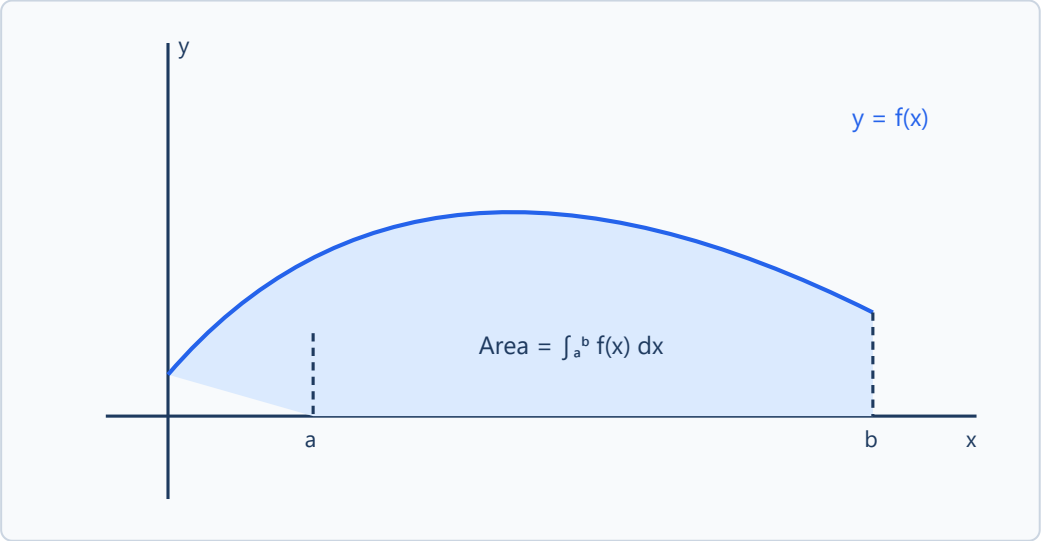
### Worked Example 3 — Evaluate $\int_1^3 (2x+1)\,dx$

```
1.  Find an antiderivative:
2.     ∫ (2x + 1) dx = x² + x    (+C, but we may drop C for definite integrals).
3.  Apply the limits:
4.     [x² + x]₁³ = (3² + 3) − (1² + 1)
5.                = (9 + 3)   − (1 + 1)
6.                = 12 − 2
7.                = 10.
8.  ∫₁³ (2x + 1) dx = 10.

// Geometric check: this integral is the area of a trapezium under y = 2x + 1
// from x = 1 to x = 3 — heights 3 and 7, width 2, area = ½(3+7)(2) = 10 ✓.
```

## 5. Area Under a Curve

If $f(x) \geq 0$ throughout $[a, b]$, the definite integral $\int_a^b f(x)\,dx$ equals the area enclosed between the curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$. If $f(x) < 0$ somewhere on the interval, the integral measures a *signed* area — portions below the axis contribute negative amounts.



**Exam tip — "area" versus "integral" can differ.** If the curve dips below the x-axis on the interval of interest, the integral gives signed area (parts below count as negative) while the *geometric area* is always positive. For geometric area you must (i) find where $f$ crosses the axis inside $[a, b]$, (ii) split the interval there, and (iii) take the absolute value of each piece before summing. Always sketch the curve first.

### Worked Example 4 — Signed vs unsigned area

Compute $\int_0^3 (x^2 - 2x)\,dx$ and also find the total geometric area between $y = x^2 - 2x$ and the x-axis on $[0, 3]$.

```
Antiderivative: ∫ (x² − 2x) dx = x³/3 − x².
Signed integral:
   [x³/3 − x²]₀³ = (9 − 9) − (0 − 0) = 0.
The signed integral is ZERO — the positive and negative parts cancel.

Find the zero of the integrand inside [0, 3]:
   x² − 2x = x(x − 2) = 0  ⇒  x = 0 or x = 2.
On (0, 2) the curve is BELOW the axis (f(1) = −1 < 0).
On (2, 3) the curve is ABOVE the axis (f(2.5) = 1.25 > 0).

Split and take absolute values:
   ∫₀² (x² − 2x) dx = (8/3 − 4) − 0 = −4/3   ⇒  |−4/3| = 4/3.
   ∫₂³ (x² − 2x) dx = (9 − 9) − (8/3 − 4) = 0 − (−4/3) = 4/3.
Total geometric area = 4/3 + 4/3 = 8/3.

// The integral is 0 but the area is 8/3 — two different numbers from the same qu
```

## 6. Area Between Curves

To find the area enclosed between two curves $y = f(x)$ (top) and $y = g(x)$ (bottom) on $[a, b]$, integrate the difference:

$$\text{Area} = \int_a^b \big[f(x) - g(x)\big]\,dx, \quad \text{where } f \geq g \text{ on } [a, b].$$

The limits $a, b$ are usually the x-coordinates of the intersection points, found by solving $f(x) = g(x)$. If the roles of "top" and "bottom" swap within the interval, split the integral at the crossover.

### Worked Example 5 — Area enclosed by $y = x^2$ and $y = 4$

```
1.  Find intersection points: x² = 4 ⇒ x = ±2.
2.  On (−2, 2) the line y = 4 is ABOVE the parabola y = x².
3.     (top − bottom) = 4 − x².
4.  Area = ∫₋₂² (4 − x²) dx
5.       = [4x − x³/3]₋₂²
6.       = (8 − 8/3) − (−8 + 8/3)
7.       = (24/3 − 8/3) − (−24/3 + 8/3)
8.       = 16/3 − (−16/3)
9.       = 32/3.
10. Area enclosed = 32/3 square units.

// By symmetry of the region in the y-axis, 2 · ∫₀² (4 − x²) dx = 2 · 16/3 = 32/3
```

## 7. Integration by Substitution

Substitution is the integration analogue of the chain rule. If a composite function $f(g(x))$ has been multiplied by the derivative $g'(x)$ of its inner function, the substitution $u = g(x)$ collapses the integrand into a pure function of $u$.

**Integration by substitution.** If $u = g(x)$ is differentiable with $du = g'(x)\,dx$, then \[ \int f(g(x))\, g'(x)\, dx = \int f(u)\, du. \] For a definite integral, the limits must also be transformed: when $x = a$, set $u = g(a)$; when $x = b$, set $u = g(b)$. Equivalently, you may substitute back to $x$ before applying the original limits.

Recognising when substitution helps: the integrand should contain a function of $g(x)$ alongside (a constant multiple of) the derivative $g'(x)$. If you spot $g'(x)$ multiplying $f(g(x))$, substitute $u = g(x)$.

## Worked Example 6 — Substitution with $u = x^2 + 1$

Find $\displaystyle\int \frac{2x}{x^2+1}\,dx$.

```
1.  Let u = x² + 1. Then du/dx = 2x, so du = 2x dx.
2.  The integrand contains 2x dx in the numerator — perfect.
3.  Rewrite:
4.     ∫ (2x / (x² + 1)) dx = ∫ (1/u) du.
5.  Integrate:
6.     ∫ (1/u) du = ln|u| + C.
7.  Substitute back:
8.     = ln|x² + 1| + C = ln(x² + 1) + C   (since x² + 1 > 0 always).

// Template: ∫ f'(x) / f(x) dx = ln|f(x)| + C — a very common pattern.
```

## Worked Example 7 — Substitution with $u = x^2$

Find $\displaystyle\int 2x\,e^{x^2}\,dx$.

```
1.  Let u = x². Then du = 2x dx.
2.  Rewrite:
3.     ∫ 2x e^(x²) dx = ∫ e^u du.
4.  Integrate:
5.     ∫ e^u du = e^u + C.
6.  Back-substitute:
7.     = e^(x²) + C.
8.  Check by differentiation:
9.     d/dx [e^(x²)] = e^(x²) · 2x ✓.
```

### Worked Example 8 — Linear substitution

Find $\int \cos(3x+1)\,dx$.

```
1.  Let u = 3x + 1. Then du = 3 dx, so dx = du/3.
2.  Rewrite:
3.     ∫ cos(3x + 1) dx = ∫ cos u · (1/3) du = (1/3) ∫ cos u du.
4.  Integrate:
5.     (1/3) sin u + C.
6.  Back-substitute:
7.     = (1/3) sin(3x + 1) + C.

// For any LINEAR inner function ax + b, the substitution just adds a factor of 1
```

**Exam tip — for definite integrals, change the limits too.** When you substitute $u = g(x)$ in $\int_a^b f(g(x))g'(x)\,dx$, the limits $a, b$ are values of $x$. Under the new variable they become $g(a), g(b)$. Either (i) change the limits and never return to $x$, or (ii) keep the original limits and substitute back to $x$ before evaluating. Mixing the two — evaluating $u$-limits on an $x$-expression — is a frequent and costly error.

## 8. Integration by Parts

Parts is the integration analogue of the product rule. Starting from $(uv)' = u'v + uv'$ and integrating both sides:

**Integration by parts.** If $u, v$ are differentiable functions of $x$, then \[ \int u\, \frac{dv}{dx}\, dx = uv - \int v\, \frac{du}{dx}\, dx, \] often written compactly as $\int u\,dv = uv - \int v\,du$. The technique trades one integral for another; for it to help, the new integral $\int v\,du$ must be simpler than the original.

Choosing $u$ and $dv$: you want $u$ to simplify on differentiation, and $dv$ to remain integrable. A standard mnemonic is **LIATE**:

$$\text{Logarithmic} \prec \text{Inverse trig} \prec \text{Algebraic} \prec \text{Trig} \prec \text{Exponential}$$

Whichever type appears earlier in this list, set it as $u$. The remaining factor becomes $dv$.

| Integrand type | Let $u =$ | Let $dv =$ | Reason |
|---|---|---|---|
| $x^n \ln x$ | $\ln x$ (L) | $x^n\, dx$ | $\ln x$ simplifies on differentiation. |
| $x^n e^{kx}$ | $x^n$ (A) | $e^{kx}\, dx$ | $x^n$ becomes a polynomial of lower degree. |
| $x^n \sin x$ or $x^n \cos x$ | $x^n$ (A) | trig $dx$ | Polynomial degree decreases. |
| $\ln x$ (alone) | $\ln x$ (L) | $dx$ | Differentiating $\ln x$ gives $1/x$. |

## Worked Example 9 — $\int x\, e^x\, dx$

```
LIATE: algebraic x before exponential e^x, so let u = x, dv = e^x dx.
Then du = dx and v = ∫ e^x dx = e^x.
Apply parts:
   ∫ x e^x dx = uv − ∫ v du
              = x e^x − ∫ e^x dx
              = x e^x − e^x + C
              = e^x (x − 1) + C.
Check: d/dx [e^x (x − 1)] = e^x (x − 1) + e^x = x e^x ✓.
```

## Worked Example 10 — $\int x \cos x\, dx$

```
LIATE: algebraic before trig. u = x, dv = cos x dx.
du = dx;  v = sin x.
Parts:
   ∫ x cos x dx = x sin x − ∫ sin x dx
                = x sin x − (−cos x) + C
```

```
                    = x sin x + cos x + C.
Check: d/dx [x sin x + cos x] = sin x + x cos x − sin x = x cos x ✓.
```

## Worked Example 11 — $\int \ln x \, dx$ (the "trick")

```
No obvious product − but write ln x = (ln x) · 1.
Let u = ln x, dv = 1 dx.
du = (1/x) dx;  v = x.
Parts:
   ∫ ln x dx = x ln x − ∫ x · (1/x) dx
             = x ln x − ∫ 1 dx
             = x ln x − x + C
             = x (ln x − 1) + C.
Check: d/dx [x ln x − x] = ln x + 1 − 1 = ln x ✓.

// The "dv = dx" trick works for any single logarithm or inverse trig.
```

## Worked Example 12 — $\int x^2 e^x \, dx$ (parts twice)

```
First pass. u = x², dv = e^x dx  ⇒  du = 2x dx, v = e^x.
   ∫ x² e^x dx = x² e^x − ∫ 2x e^x dx
               = x² e^x − 2 ∫ x e^x dx.
Use Worked Example 9: ∫ x e^x dx = e^x (x − 1) + C₁.
Substitute:
   ∫ x² e^x dx = x² e^x − 2 [ e^x (x − 1) ] + C
               = x² e^x − 2x e^x + 2 e^x + C
               = e^x (x² − 2x + 2) + C.
Check: d/dx [e^x (x² − 2x + 2)] = e^x (x² − 2x + 2) + e^x (2x − 2)
   = e^x [x² − 2x + 2 + 2x − 2] = x² e^x ✓.

// Rule of thumb: ∫ xⁿ e^(kx) dx needs parts applied n times.
```

**Exam tip — LIATE is a guideline, not a law.** It works in the vast majority of A-level integrals, but if LIATE gives an integrand that does not simplify, try the other way. The real test is whether $\int v\,du$ is simpler than $\int u\,dv$. Also, for integrands like $\int x \ln x\,dx$, LIATE sends you to $u = \ln x$ (correct), but always double-check by trying to integrate the alternative to see if it terminates.

## 9. Integration Using Partial Fractions

A rational function $P(x)/Q(x)$ (with $\deg P < \deg Q$) can often be decomposed into a sum of simpler fractions. Each piece integrates to a logarithm or power. For distinct linear factors in the denominator:

$$\frac{f(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}, \qquad \int \frac{1}{x-a}\,dx = \ln|x-a| + C.$$

### Worked Example 13 — Partial fractions

Find $\displaystyle\int \frac{3x+5}{(x-1)(x+2)}\,dx.$

```
1.  Write: (3x + 5) / [(x − 1)(x + 2)] = A/(x − 1) + B/(x + 2).
2.  Multiply through:
3.     3x + 5 = A(x + 2) + B(x − 1).
4.  Cover-up at x = 1:   3(1) + 5 = A(3)   ⇒  8 = 3A  ⇒  A = 8/3.
5.  Cover-up at x = −2:  3(−2) + 5 = B(−3) ⇒ −1 = −3B ⇒  B = 1/3.
6.  Rewrite the integrand:
7.     (3x + 5) / [(x − 1)(x + 2)] = (8/3)/(x − 1) + (1/3)/(x + 2).
8.  Integrate term by term:
9.     ∫ ... dx = (8/3) ln|x − 1| + (1/3) ln|x + 2| + C.

// Always write the cover-up values at the "bad" points of the denominator.
// Combine the two logs if asked: = (1/3) ln|(x − 1)⁸ (x + 2)| + C.
```

## 10. Integration Using Trigonometric Identities

Many trigonometric integrands don't match the standard forms directly, but they can be rewritten using identities. The most useful identities for integration are:

| Identity | Useful for |
|---|---|
| $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ | $\int \cos^2 x \, dx$ |
| $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ | $\int \sin^2 x \, dx$ |
| $\sin 2x = 2 \sin x \cos x$ | $\int \sin x \cos x \, dx$ |
| $\sec^2 x = 1 + \tan^2 x$ | $\int \tan^2 x \, dx$ |
| Product-to-sum, e.g. $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$ | $\int \sin(mx)\cos(nx) \, dx$ |

## Worked Example 14 — $\int \cos^2 x \, dx$

```
1.  Apply the double-angle identity cos²x = (1 + cos 2x)/2.
2.  Rewrite:
3.     ∫ cos²x dx = ∫ ½(1 + cos 2x) dx
4.                = ½ ∫ 1 dx + ½ ∫ cos 2x dx
5.                = ½ x + ½ · ½ sin 2x + C
6.                = x/2 + (sin 2x)/4 + C.
7.  Check: d/dx [x/2 + (sin 2x)/4] = 1/2 + (cos 2x)/2 = (1 + cos 2x)/2 = cos²x ✓.

// The same trick with sin²x = (1 − cos 2x)/2 gives ∫ sin²x dx = x/2 − (sin 2x)/4
```

## Worked Example 15 — Product-to-sum

Find $\displaystyle\int \sin(2x)\cos(3x)\,dx$.

```
1.  Apply product-to-sum:
2.     sin A cos B = ½ [sin(A + B) + sin(A − B)].
3.     sin(2x) cos(3x) = ½ [sin(5x) + sin(−x)]
```

```
                    = ½ sin(5x) − ½ sin x.        (sin(−x) = −sin x)
Integrate:
    ∫ sin(2x) cos(3x) dx = ∫ [½ sin(5x) − ½ sin x] dx
                         = ½ · (−1/5) cos(5x) − ½ · (−cos x) + C
                         = −(1/10) cos(5x) + ½ cos x + C.
```

**Exam tip — cos²x and sin²x cannot be integrated directly.** Students often try $\int \cos^2 x\,dx = \frac{\cos^3 x}{3}$ (applying the power rule as if $\cos x$ were $x$) — this is wrong, because differentiating $\cos^3 x/3$ gives $-\cos^2 x \sin x$, not $\cos^2 x$. The only A-level-safe method is to linearise with the double-angle identity first.

## 11. Volumes of Revolution

Rotating the region between a curve $y = f(x)$, the x-axis, and the lines $x = a$, $x = b$ through a full turn about the x-axis sweeps out a solid whose volume can be computed by integration. The thin-disk approximation: each thin strip of width $dx$ at position $x$ becomes a disk of radius $y = f(x)$ and thickness $dx$, contributing volume $\pi y^2\,dx$. Integrating gives:

**Solid of revolution — volume formula.** The solid obtained by rotating the curve $y = f(x)$ around the x-axis from $x = a$ to $x = b$ has volume \[ V = \pi \int_a^b y^2\, dx = \pi \int_a^b \big[f(x)\big]^2\, dx. \] For rotation about the y-axis of the curve $x = g(y)$ from $y = c$ to $y = d$, \[ V = \pi \int_c^d x^2\, dy = \pi \int_c^d \big[g(y)\big]^2\, dy. \]

| Axis of revolution | Variable of integration | Volume formula |
|---|---|---|
| x-axis | $x$ | $V = \pi \int_a^b y^2\,dx$ |
| y-axis | $y$ | $V = \pi \int_c^d x^2\,dy$ |

### Worked Example 16 — Volume of revolution of $y = \sin x$ about the x-axis, $0 \le x \le \pi$

```
Apply the disk formula:
   V = π ∫₀^π y² dx = π ∫₀^π sin²x dx.
Use sin²x = (1 − cos 2x)/2:
   V = π ∫₀^π ½(1 − cos 2x) dx
     = (π/2) [x − (sin 2x)/2]₀^π.
Evaluate:
   at x = π:  π − (sin 2π)/2 = π − 0 = π.
   at x = 0:  0 − (sin 0)/2  = 0.
V = (π/2) · π = π²/2.

// Volume of the "lemon" obtained by rotating one arch of a sine about the x-axis
// Numerical: π²/2 ≈ 4.9348 cubic units.
```

## 12. First-Order Separable Differential Equations

A differential equation relates a function to its derivative(s). The simplest useful case is a *first-order separable* equation in which the derivative factors into a function of $x$ times a function of $y$:

**First-order separable differential equation.** An equation of the form \[ \frac{dy}{dx} = f(x)\, g(y) \] is called *separable*. Separate the variables by treating $dy/dx$ as a quotient of differentials: \[ \frac{1}{g(y)}\, dy = f(x)\, dx \quad\Longrightarrow\quad \int \frac{1}{g(y)}\, dy = \int f(x)\, dx + C. \] The resulting equation (after integrating both sides) is called the *general solution*. An *initial condition* $y(x_0) = y_0$ pins $C$ down to give the *particular solution*.

### Worked Example 17 — Solve $dy/dx = 2y$ with $y(0) = 3$

```
Separate variables (valid while y ≠ 0):
   (1/y) dy = 2 dx.
Integrate both sides:
   ∫ (1/y) dy = ∫ 2 dx
   ln|y| = 2x + C.
Exponentiate:
   |y| = e^(2x + C) = e^C · e^(2x).
   y = A · e^(2x)        (A = ±e^C, any nonzero constant.)
```

```
Apply the initial condition y(0) = 3:
   3 = A · e^0 = A  ⇒  A = 3.
Particular solution: y = 3 e^(2x).
Check: dy/dx = 6 e^(2x) = 2 · 3 e^(2x) = 2y ✓. y(0) = 3 ✓.

// dy/dx proportional to y gives exponential growth — the classic population /
// radioactive-decay model.
```

### Worked Example 18 — Solve $dy/dx = xy$ with $y(0) = 1$

```
Separate:
   (1/y) dy = x dx.
Integrate:
   ∫ (1/y) dy = ∫ x dx
   ln|y| = x²/2 + C.
Exponentiate:
   y = A · e^(x²/2).
Initial condition y(0) = 1:
   1 = A · e^0  ⇒  A = 1.
Particular solution:  y = e^(x²/2).
Check: dy/dx = e^(x²/2) · x = x y ✓.
```

## 13. Applications in Mechanics (SUVAT Links)

Integration provides the link between acceleration, velocity, and displacement for a particle in motion along a line. If $a(t)$ is the acceleration as a function of time:

$$v(t) = \int a(t)\,dt + C_1, \qquad s(t) = \int v(t)\,dt + C_2.$$

The constants $C_1, C_2$ are fixed by initial conditions (usually $v(0) = u$ the initial velocity, and $s(0) = 0$ the starting position). For constant acceleration $a$, integrating gives the familiar SUVAT equations:

$$v = u + at, \qquad s = ut + \tfrac{1}{2}at^2.$$

The power of integration is that it handles variable acceleration too. If $a(t) = 6 - 2t$ and a particle starts from rest, then $v(t) = 6t - t^2$ and $s(t) = 3t^2 - t^3/3$ — no SUVAT formula applies, but integration still gives the answer immediately.

## 14. Applied Worked Examples

### Worked Example 19 — $\int x \ln x \, dx$ (parts with LIATE)

```
LIATE: logarithmic before algebraic, so u = ln x, dv = x dx.
du = (1/x) dx;  v = x²/2.
Parts:
   ∫ x ln x dx = (x²/2) ln x − ∫ (x²/2)(1/x) dx
               = (x²/2) ln x − ∫ (x/2) dx
               = (x²/2) ln x − x²/4 + C
               = (x²/4)(2 ln x − 1) + C.
Check: d/dx [(x²/2) ln x − x²/4] = x ln x + x²/2 · (1/x) − x/2
                                 = x ln x + x/2 − x/2 = x ln x ✓.
```

## 15. Summary — When to Use Which Technique

| Integrand pattern | Technique | Key formula |
|---|---|---|
| $x^n$ or polynomial | Power rule | $\int x^n \, dx = x^{n+1}/(n+1) + C$ |
| Standard trig / exp / log | Quote standard integral | See table in §3. |
| $f(g(x)) \cdot g'(x)$ | Substitution $u = g(x)$ | $\int f(u) \, du$ |

| | | |
|---|---|---|
| Product of different function types (algebraic × exp / trig / log) | Integration by parts, with LIATE | $\int u\,dv = uv - \int v\,du$ |
| Rational function $P(x)/Q(x)$ ($\deg P < \deg Q$) | Partial fractions | Decompose, integrate each piece to $\ln$. |
| $\sin^2, \cos^2, \sin mx \cos nx$ | Trig identity (linearise) | Double-angle or product-to-sum. |
| Area between $f$ and $g$ | Definite integral of difference | $\int_a^b (f-g)\,dx$ |
| Volume of revolution about x-axis | Disk method | $V = \pi \int_a^b y^2\,dx$ |
| $dy/dx = f(x)g(y)$ | Separate and integrate | $\int dy/g(y) = \int f(x)\,dx + C$ |

**Chapter-wide checklist for Integration.**

- **Always write** $+C$ on every indefinite integral.
- **For definite integrals, no** $C$ **— but always show** $[F(x)]_a^b$ and the subtraction $F(b) - F(a)$.
- **Substitution: change the limits** if it's a definite integral, or substitute back if indefinite.
- **Parts: use LIATE** to pick $u$. Check whether $\int v\,du$ is actually simpler.
- **Sign of area:** if the curve dips below the axis, split the integral at the zeros.
- **For area between curves:** find intersection points first, then (top − bottom).
- $\cos^2 x$ **and** $\sin^2 x$**:** always linearise with the double-angle formula before integrating.
- **Volumes of revolution:** it is $y^2$, not $y$ — remember to square $f(x)$.
- **Differential equations:** after integrating, always apply the initial condition to fix $C$.
- **Check by differentiation** whenever time permits — a 10-second sanity check.

**Typical OCR examiner complaints (from mark reports).**

- Forgetting $+C$ in indefinite integrals — a reliable way to lose an easy mark.

- Changing the variable via substitution but forgetting to change the limits, or evaluating $u$-limits on an $x$-expression.
- Choosing $u$ and $dv$ the wrong way round in integration by parts and then getting stuck in a harder integral.
- Integrating $\cos^2 x$ as $\cos^3 x/3$ (wrong — not a power of $x$).
- In volumes of revolution, integrating $y$ instead of $y^2$, or forgetting the factor of $\pi$.
- For "area between a curve and the x-axis", failing to check whether the curve crosses the axis — taking the integral alone when the answer required the geometric area.
- In partial fractions, cover-up arithmetic errors, or decomposing a non-proper rational function (where $\deg P \geq \deg Q$) without first doing polynomial division.
- Solving a differential equation but forgetting to apply the initial condition — leaving $C$ unspecified when the question demanded the particular solution.

**Formula snapshot (memorise).**

- Power rule: $\int x^n\,dx = x^{n+1}/(n+1) + C$ (for $n \neq -1$). $\int 1/x\,dx = \ln|x| + C$.
- Trig: $\int \sin x\,dx = -\cos x + C$; $\int \cos x\,dx = \sin x + C$; $\int \sec^2 x\,dx = \tan x + C$.
- Exp: $\int e^x\,dx = e^x + C$; $\int e^{kx}\,dx = (1/k)e^{kx} + C$.
- Linear inside: $\int f(ax+b)\,dx = (1/a)F(ax+b) + C$.
- Definite: $\int_a^b f(x)\,dx = F(b) - F(a)$.
- Substitution: $u = g(x)$, $du = g'(x)\,dx$. Change limits for definite.
- Parts: $\int u\,dv = uv - \int v\,du$. Choose $u$ by LIATE.
- Partial fractions: decompose, then each piece integrates to $\ln$ or a power.
- $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$; $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$.
- Volume of revolution: $V = \pi \int_a^b y^2\,dx$ (about x-axis); $V = \pi \int_c^d x^2\,dy$ (about y-axis).
- Separable ODE: $\int dy/g(y) = \int f(x)\,dx + C$; apply initial condition.